

1. SCOPE

This Standard establishes the general permissible tolerances for sheet metal and milled parts.

2. APPLICATION

This Standard applies to parts manufactured for assembly on Embraer-made aircraft.

3. APPLICABLE DOCUMENTS

The following documents form a part of this Standard and shall be complied with to the extent herein required.

Embraer Engineering Standards:

- NE 03-006: Riveting-Drilling, Countersinking and Dimpling.
- NE 03-008: Countersinking and Dimpling of holes for Bolts and Screws.
- NE 03-010: Permissible Discrepancies in Milling Operations.
- NE 06-003: Bolts, Screws, Nuts and Washers - Installation.

Drafting Room Manual (MED)

- Section 6.5: Bolts and Screws.

4. GENERAL CONDITIONS

4.1 GENERAL TOLERANCES

The general tolerances established by this Standard are compatible with usual manufacturing practices, that is:

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16	E	17	C	18	D		

STANDARD APPROVED BY EO N/ 063274 XNT

DPR _____ APPROVED
DTE _____

SHEET METAL AND MILLED PARTS - TOLERANCES

NE 03-007

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$\pm 1^0$ for angular dimensions.

± 0.8 mm for dimensions obtained through sheet forming work.

± 0.3 mm for dimensions obtained through milling.

4.2 USE OF GREATER TOLERANCES

Tolerances greater than those hereby recommended must always be specified in the drawings, as long as they do not affect the part, assembly or function of adjoining parts.

Therefore, for external or internal dimensions of parts manufactured through sheet metal forming, the tolerance values may be increased from ± 0.8 mm to ± 1.6 mm, or the tolerances for the angular dimensions changed to $\pm 2^0$, etc.

Whenever these changes are possible, they shall be clearly indicated on the drawing, beside the nominal dimensions to which the tolerance applies.

A case in point is seen in Figure 18, item 6.1.3.

4.3 BALL JOINTS AND BEARINGS

Tolerances for pins or ball joint and bearing housings must always be indicated on the drawing, and according to the values recommended in the manufacturer's manuals.

4.4 ORDER OF MAGNITUDE

The number of digits of a tolerance value indicated for a given dimension shall be equal to the number of digits to the right of the point, in the nominal dimension.

Table 1 below exemplifies the magnitude order of tolerances in a few cases.

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TABLE 1 : EXAMPLES OF TOLERANCE MAGNITUDE ORDER

DIMENSION (mm)	TOLERANCES (mm)	
	CORRECT	INCORRECT
15, <u>25</u>	0, <u>30</u>	0,3
7, <u>80</u>	0, <u>80</u>	0,825
8, <u>3</u>	0, <u>5</u>	0,50
15, <u>0</u>	0, <u>3</u>	0,30
15, <u>00</u>	0, <u>75</u>	0,7

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5. SPECIFIC CONDITIONS

5.1 GENERAL TOLERANCES FOR FORMED PARTS

5.1.1 LINEAR AND ANGULAR DIMENSIONS

Sheet parts formed by bending, swaging, hand-lathing, milling on Router machine, shall be produced with general tolerance of ± 0.8 mm for linear dimensions and $\pm 1^\circ$ for angular dimensions. See Figure 1a.

EXCEPTIONS

- tolerance of flange height of parts formed on hydroform or rubber press:
 $+ 1.4$
 $- 1.0$ mm ;
- tolerance of all linear dimensions for parts with all surfaces class IV (NE 03-040): 2 mm;
- tolerance of bending radius R: $+ 0.8$
 0 mm ;
- tolerance of joggles L x S: $+ 0.8$
 0 mm.

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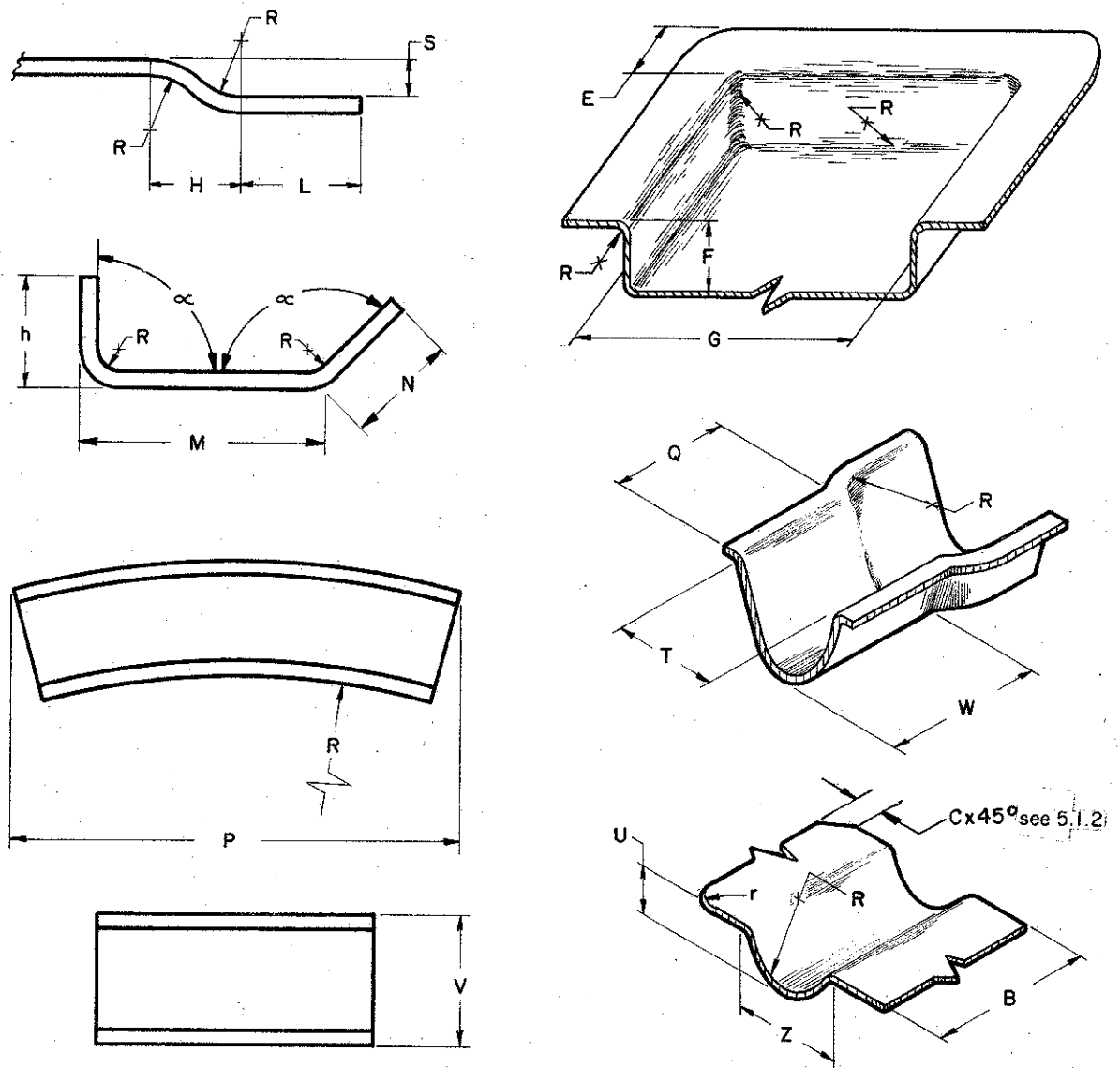


FIGURE 1a: DIMENSIONS SUBJECT TO GENERAL TOLERANCES

5.1.2 SHOP CORNER OPTIONS OF SHEET PARTS

If drawing does not indicate the note "NL XXX : CORNER MUST BE AS SHOWN", shop may change corners according to options specified in Figure 1b.

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SHOWN ON DRAWING	SHOP OPTION (S)	

FIGURE 1b : SHOP CORNER OPTIONS

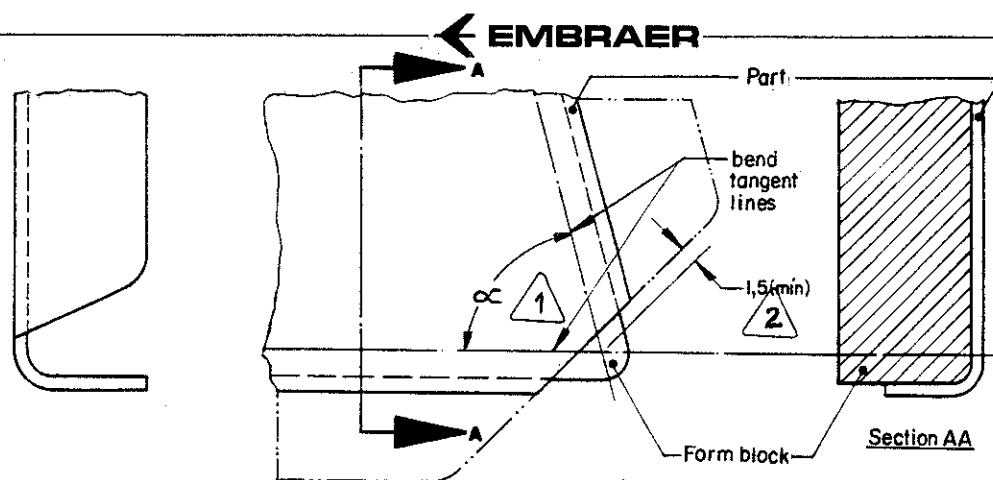
5.1.3 RADII RELIEF

Although other way specified on drawing, sheet parts containing corners formed by two convergent flanges presenting problems upon forming such as bending radius smaller than the minimum specified value, cracks, superficial deformations, etc shall present a radius relief according to one of the configurations specified in Figure 2.

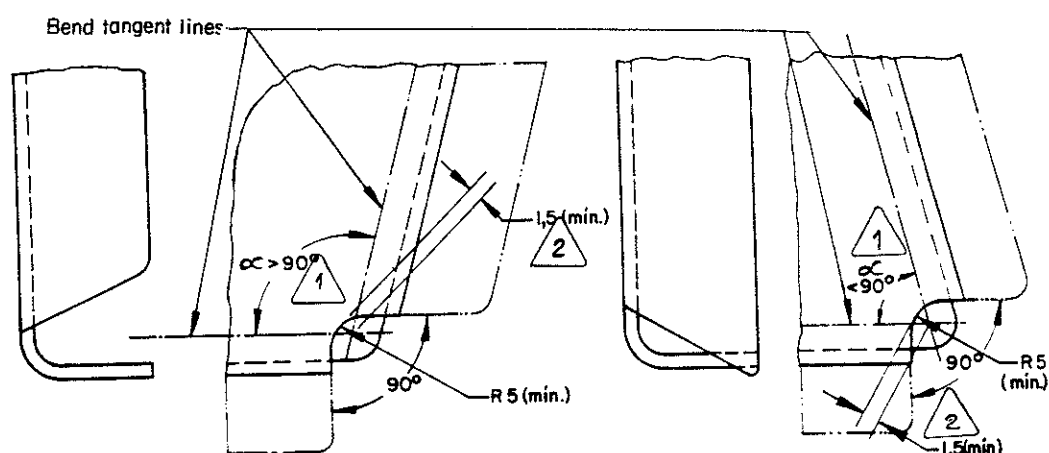
NOTES

- 1) Radius relief shall not interfere in the functional performance of part, such as reduction of edge distance, reduction of mechanical strength, etc. In case of doubt, Structural Engineering shall be consulted.
- 2) Class IV parts according to NE 03-040 do not require radius relief.

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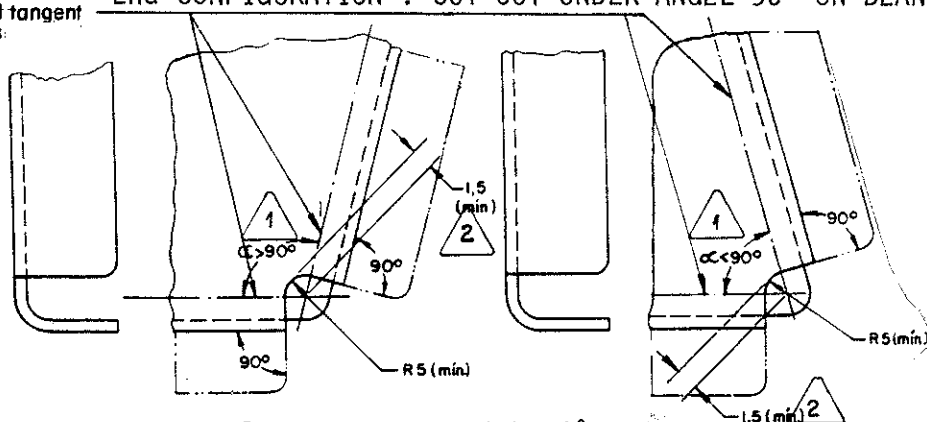


1st CONFIGURATION : STRAIGHT CUT OUT ON BLANK



NOTE

It shall be considered, in this case, the final position of bent flange if the angle α is different from 90° .

2nd CONFIGURATION : CUT OUT UNDER ANGLE 90° ON BLANK

NOTE

This cut out promotes flange corners with 90° , independent of angle α .

3rd CONFIGURATION : PERPENDICULAR CUT OUT TO THE BEND TANGENT LINES ON BLANK

- 1 Angle α may be equal or different from 90° for all three preferences.
- 2 The maximum value (≈ 2.5 mm) is free, since it does not interfere on functional performance of parts.

FIGURE 2 : RADIUS RELIEF

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5.1.4 STRAIGHTNESS AND FLATNESS TOLERANCES

Straightness tolerances of flanges and other similar details shall be according to Figure 3a.

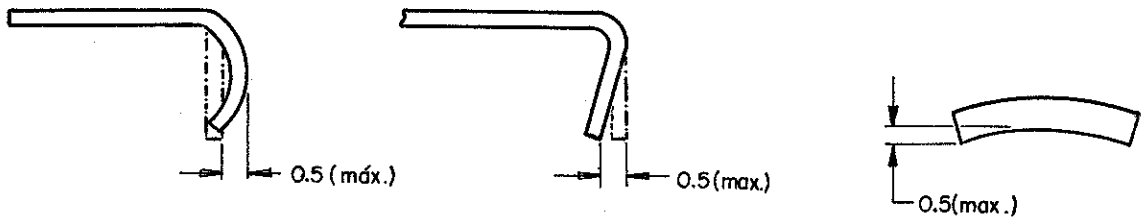


FIGURE 3a : TOLERANCE FOR STRAIGHTNESS OF PARTS

The tolerance for web flatness in parts formed through drawing shall be within 0.8 mm maximum. See Figure 3b.

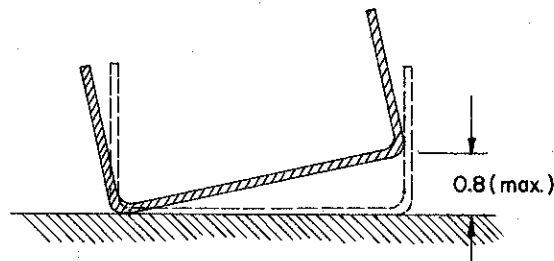
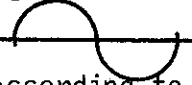


FIGURE 3b : TOLERANCE FOR WEB FLATNESS

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5.1.5 LUGS CUTOUT

a) General Rule

The symbol  in lug representation on drawings means that it shall be cutout according to part shape. See Figure 3c.

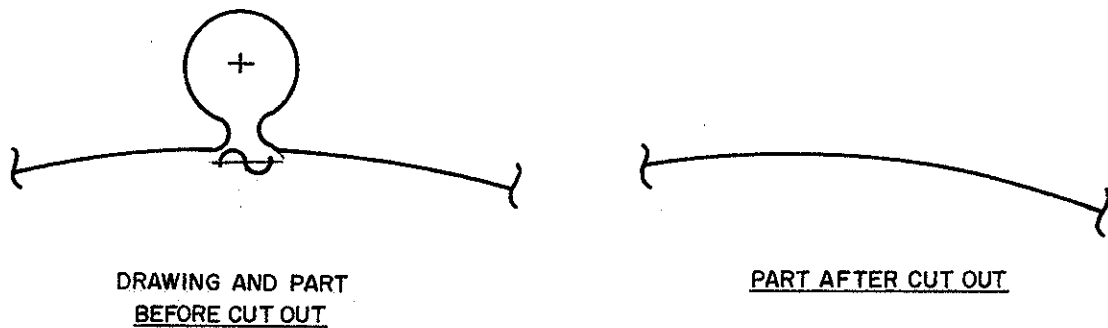


FIGURE 3c : LUG CUTOUT, GENERAL RULE

b) Lug on Winding Shape

In this case the lug cutout shall be always straight, according to indicated on Figure 3d.

The part is authorized to have different contour from that indicated on drawing, in the lug (cutout) region.

b1) Lug Indicated on Drawing

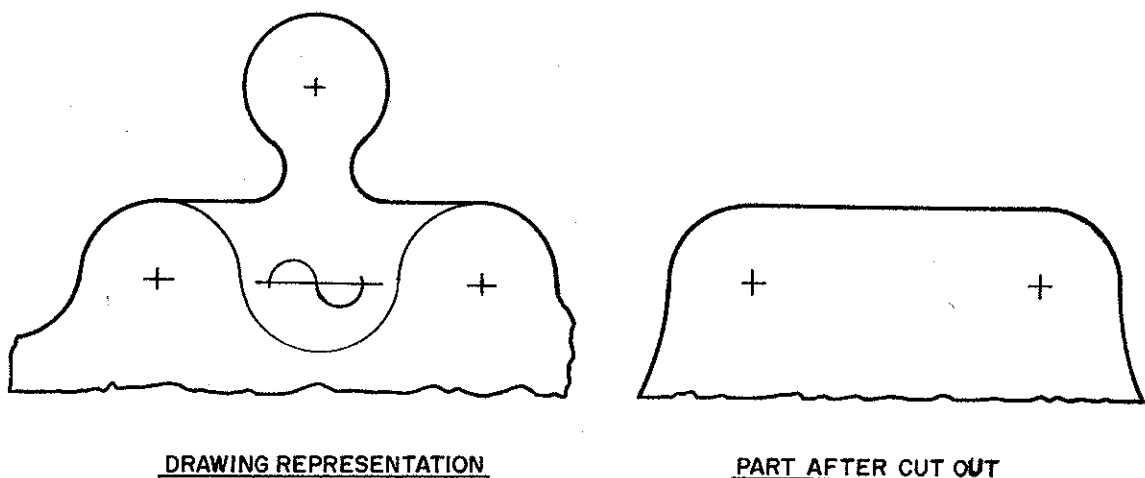
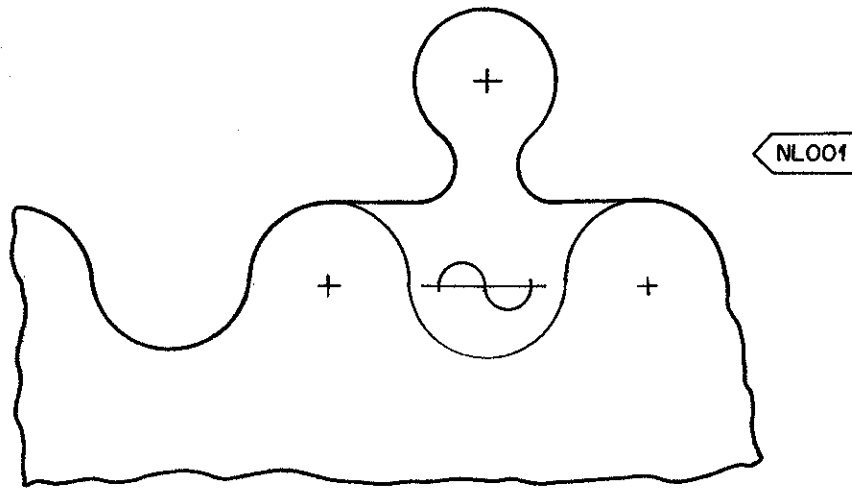


FIGURE 3d : LUG INDICATED ON DRAWING

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NOTE

If the winding shape must remain after a lug cutout, a free note (NL XXX) shall be added on drawing. See example in Figure 3e.



NL 001 - The winding shape must remain after lug cutout.

FIGURE 3e : EXAMPLE OF FREE NOTE INDICATION FOR WINDING SHAPE REMAIN

b2) Lug Not Indicated on Drawing

In this case the lug is only made to facilitate the manufacturing process and is added to the part according to production necessity.

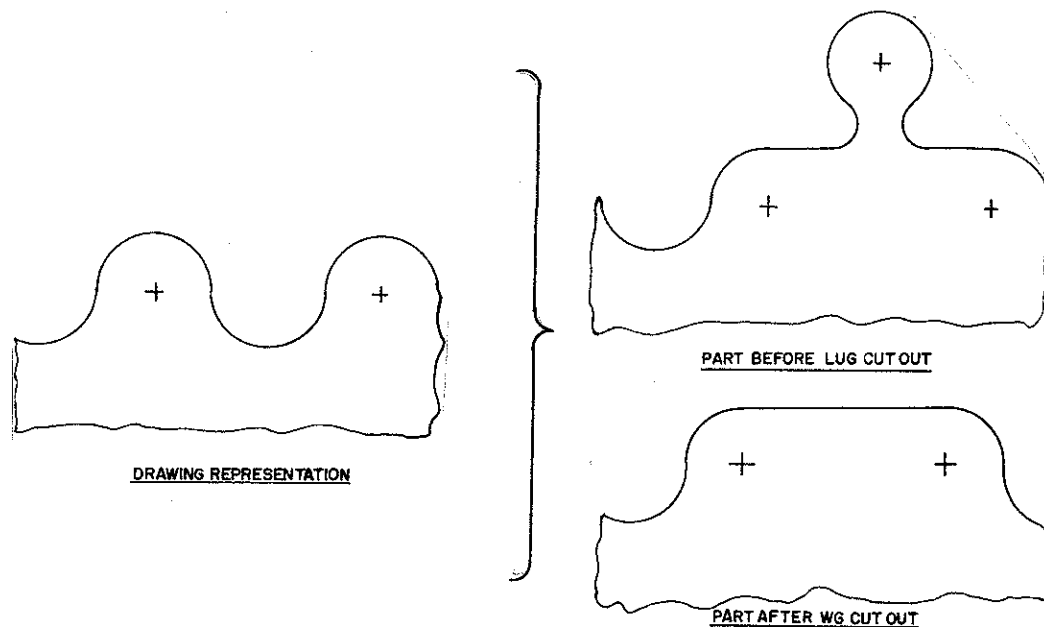


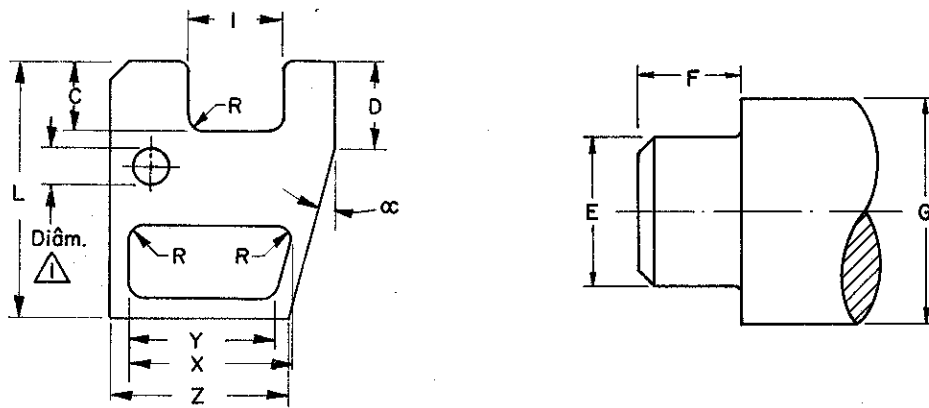
FIGURE 3f : LUG NOT INDICATED ON DRAWING

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5.2 TOLERANCES FOR MECHANICALLY MILLED PARTS

5.2.1 NOMINAL DIMENSIONS

The tolerances for nominal dimensions, external or internal, in mechanically milled parts (cutting, turning, etc.) shall be within a ± 0.3 mm limit, unless the drawing indicates otherwise. See Figure 4.



- 1 When the tolerances for the holes can or are to be different from those specified in item 5.4.3, they must be clearly indicated in the drawings.

FIGURE 4 : TOLERANCES FOR NOMINAL DIMENSIONS

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5.2.4 DISCREPANCIES IN CUTTING OPERATIONS

The tolerances for discrepancies in cutting operations shall be in accordance with NE 03-010.

5.2.5 THICKNESS OF MILLED WALLS

The tolerance for milled walls is ± 0.3 mm -0.0 mm.

Figure 7, below, shows the dimensions of a part with milled walls (for instance, a rib).

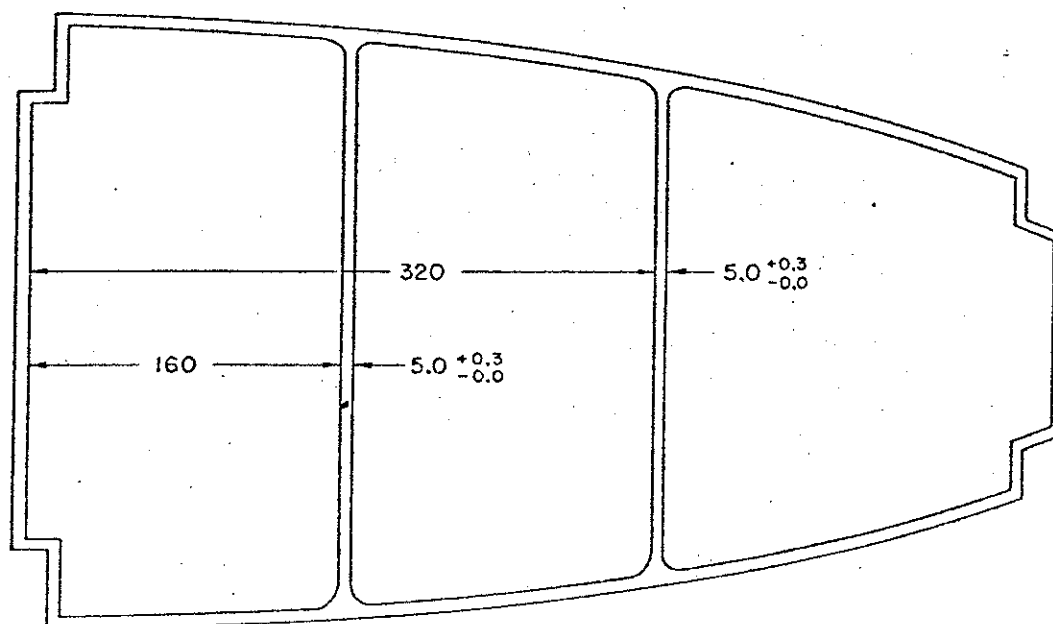


FIGURE 7: TOLERANCE FOR MILLED WALL THICKNESS

5.3 TOLERANCES FOR CHEMICALLY MILLED PARTS

5.3.1 TOLERANCES FOR CUTTING DEPTH

The tolerances for cutting depth in chemically milled aluminum alloy parts are represented in Table 2 by the tolerance for thickness of remaining material, valid when good production practices are used.

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TABLE 2 : TOLERANCES FOR CUTTING DEPTH

CUTTING DEPTH		TOLERANCE ($\pm$) FOR REMAINING MATERIAL THICKNESS	
IN	mm	IN	mm
UP TO 0.156	UP TO 3.96	0.002	0.06
0.157 thru 0.250	3.97 thru 6.35	0.003	0.08
0.251 thru 0.312	6.36 thru 7.93	0.004	0.10
0.313 thru 0.375	7.94 thru 9.52	0.005	0.13
0.376 thru 0.500	9.53 thru 12.70	0.006	0.15

NOTE: The following values should be added to the tolerances listed above:
 $\pm$ 0.001 in (0.03 mm) if a drawing operation is carried out before the chemical milling, $\pm$ 0.005 in (0.13 mm) if a mechanical milling operation is carried out before the chemical milling, and $\pm$ 0.015 in (0.38 mm) if the part is forged.

5.3.2 LATERAL TOLERANCES

The lateral tolerances for chemically milled aluminum alloy parts are listed in Table 3. They are valid when good production practices are used.

TABLE 3 : LATERAL TOLERANCES

CUTTING DEPTH		LATERAL TOLERANCE	
IN	mm	IN	mm
UP TO 0.300	UP TO 7.62	0.030	0.8
0.301 thru 0.350	7.63 thru 8.89	0.031	0.8
0.351 thru 0.400	8.90 thru 10.16	0.035	0.9
0.401 thru 0.450	10.17 thru 11.43	0.040	1.0
0.451 thru 0.500	11.44 thru 12.70	0.050	1.3

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Figure 8 below shows the effectivity of lateral tolerances in a chemically milled part.

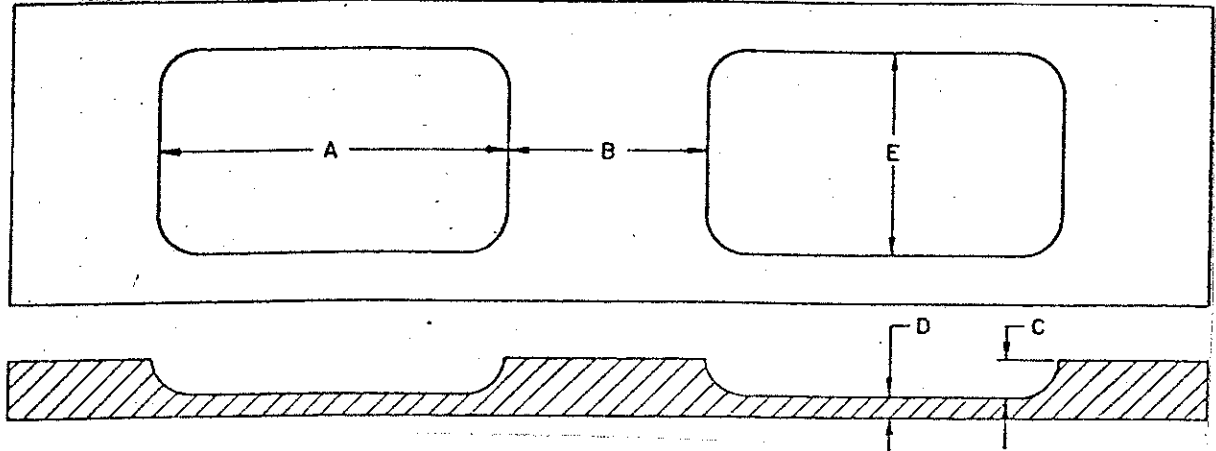


FIGURE 8: LATERAL TOLERANCES EFFECTIVITY

If, for instance, the cutting depth (C) is 0.050 in (1.27 mm), the lateral tolerance, for dimensions (A, B and E) as per Table 3, shall be ± 0.030 in (0.8 mm).

On the other hand, the tolerance for cutting depth will be known by means of the remaining material thickness tolerance (D), and shall be determined as per item 5.3.1.

5.4 TOLERANCES FOR DRILLING, COUNTERSINKING, AND DIMPLING

5.4.1 TOLERANCES FOR RIVETING

5.4.1.1 DRILLING, COUNTERSINKING, DIMPLING

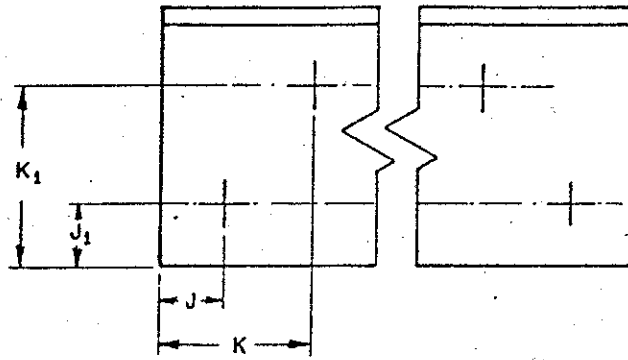
The tolerances for drilling, countersinking, and dimpling in riveting operations are established in chapter 4, NE 03-006.

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5.4.1.2 DISTANCE BETWEEN RIVET CENTER AND SHEET EDGES

The tolerances for the distance between the rivet hole center and the edge of the sheet, unless otherwise specified in the drawings, shall be in accordance with Figure 9, below.

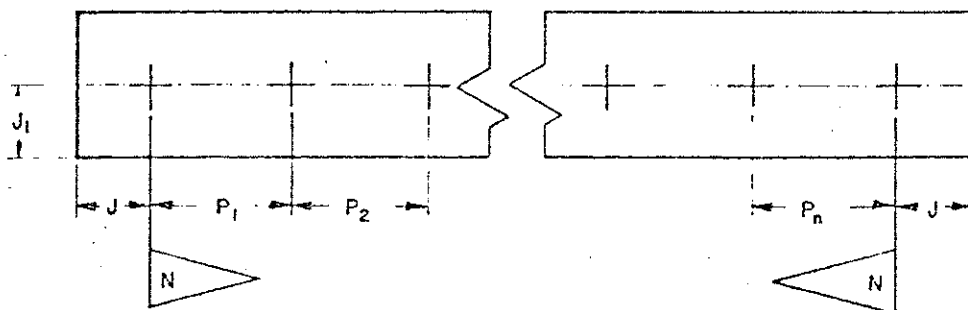


Tolerance for dimension J or $J_1 = \pm 0.8 \text{ mm}$
 Tolerance for dimension K or $K_1 = \pm 0.8 \text{ mm}$

FIGURE 9: TOLERANCE FOR THE DISTANCE BETWEEN HOLE CENTER AND EDGE

5.4.1.3 DISTANCE BETWEEN HOLE CENTERS

Whenever the rivet drilling is indicated by flags, the pitch (distance between the hole centers) shall not vary by more than $\pm 1 \text{ mm}$. See Figure 10.



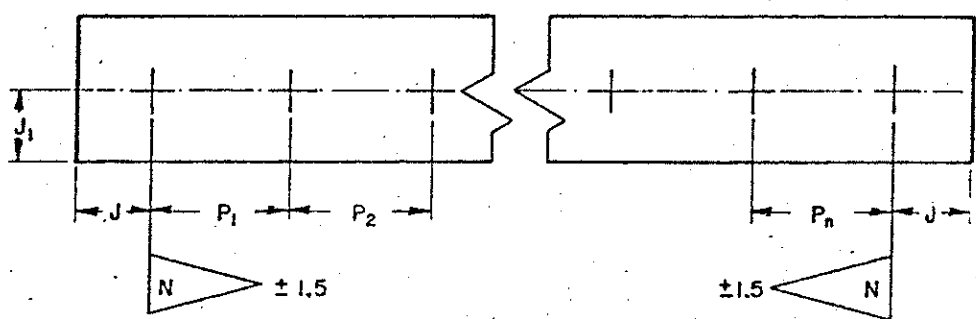
P = Distance between hole centers = Riveting pitch
 $P_1 - P_2 = \pm 1 \text{ mm (max)}$ pr $P_1 - P_n = \pm 1 \text{ mm (max)}$, etc.

FIGURE 10: RIVETING PITCH TOLERANCES

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If, on the other hand, the design requires other pitch limits, that is, different from ± 1 mm, they should be indicated opposite to the flag, as shown in figure 11.



$P_1 - P_2 = \pm 1.5 \text{ mm (max)}$ or $P_1 - P_n = \pm 1.5 \text{ mm (max)}$, etc.

FIGURE 11: INDICATION OF OTHER RIVETING PITCH TOLERANCES

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5.4.2 BOLT HOUSING

5.4.2.1 DRILLING

The tolerances for hole dimensions in bolt attachment are established in Table 10, item 6.2.3 MED. These tolerances shall be indicated in the drawing.

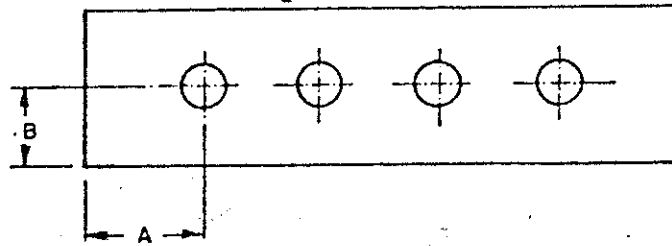
5.4.2.2 COUNTERSINKING AND DIMPLING

The tolerances for countersinking and dimpling, for bolt housing, are established in Chapter 4, NE 03-008.

5.4.2.3 DISTANCE BETWEEN THE HOLE CENTERS AND PART EDGES

When drilling bolt holes, the tolerances for the distances between the hole centers and the part edges shall be within ± 0.5 mm, unless otherwise specified in the drawing. See Figure 12.

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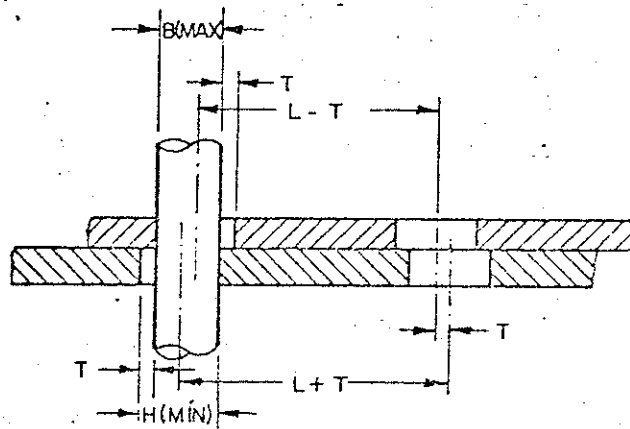


Tolerance for B, A = ± 0.5 mm

FIGURE 12: TOLERANCE FOR THE DISTANCE BETWEEN HOLE CENTER AND EDGE

5.4.2.4 DISTANCE BETWEEN HOLES

When drilling bolt holes, the tolerances should be listed in the drawings. In figure 13, below, this specification is shown, considering hole pitch (L) tolerance equal to $\pm T$.



Min clearance = H min - B max = A
Tolerance = H min - B max = T

$T = A$, that is, the tolerance for the space between holes should be equal to the minimum clearance between bolt and hole. This is an example for two superimposed holes; for more than two holes use $T = 0.35A$.

FIGURE 13: TOLERANCE FOR THE DISTANCE BETWEEN BOLT HOLES

5.4.2.5 HOLES FOR SELF-TAPPING SCREWS

The tolerance for hole dimensions in self-tapping screw attachments are established in Table 7, NE 06-003. These tolerances shall be indicated in the drawing.

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5.4.3 HOLES LISTED ACCORDING TO THEIR NOMINAL DIMENSIONS

The tolerances for holes whose listing on the drawings is based on their nominal diameters only, must be in accordance with Table 4.

TABLE 4 : TOLERANCES FOR HOLES LISTED ACCORDING TO THEIR NOMINAL DIMENSIONS

DRILL BIT	NOMINAL DIMENSION ¹ / ₂		HOLE TOLERANCE		DRILL BIT	NOMINAL DIMENSION ¹ / ₂		HOLE TOLERANCE	
	in	mm	in	mm		in	mm	in	mm
62	0,0380	1,0	0,037 0,042	0,9 1,1	3/8	0,3750	9,5	0,374 0,381	9,5 9,7
50	0,0700	1,8	0,069 0,074	1,8 1,9	Y	0,3770	9,6	0,376 0,383	9,5 9,7
48	0,0760	1,9	0,075 0,080	1,9 2,0	25/64	0,3906	9,9	0,390 0,397	9,9 10,1
5/64	0,0781	2,0	0,077 0,082	2,0 2,1	7/16	0,4375	11,1	0,437 0,444	11,1 11,3
45	0,0820	2,1	0,081 0,086	2,1 2,2	29/64	0,4531	11,5	0,452 0,459	11,5 11,7
3/32	0,0938	2,4	0,093 0,098	2,4 2,5	15/32	0,4688	11,9	0,468 0,475	11,9 12,1
36	0,1065	2,7	0,105 0,110	2,7 2,8	1/2	0,5000	12,7	0,499 0,506	12,7 12,9
31	0,1200	3,0	0,119 0,124	3,0 3,1	33/64	0,5156	13,1	0,515 0,524	13,1 13,3
1/8	0,1250	3,2	0,124 0,129	3,1 3,3	17/32	0,5312	13,5	0,530 0,539	13,5 13,7
29	0,1360	3,4	0,135 0,141	3,4 3,6	37/64	0,5781	14,7	0,577 0,586	14,7 14,9
9/64	0,1406	3,6	0,139 0,145	3,5 3,7	19/32	0,5938	15,1	0,593 0,602	15,1 15,3
26	0,1470	3,7	0,146 0,152	3,7 3,9	5/8	0,6250	15,9	0,624 0,633	15,9 16,1
19	0,1660	4,2	0,166 0,171	4,2 4,3	41/64	0,6406	16,3	0,640 0,649	16,3 16,5
17	0,1730	4,4	0,172 0,178	4,4 4,5	21/32	0,6562	16,7	0,655 0,664	16,6 16,9
16	0,1770	4,5	0,176 0,182	4,5 4,6	3/4	0,7500	19,0	0,749 0,758	19,0 19,3
11	0,1910	4,8	0,190 0,196	4,8 5,0	49/64	0,7656	19,4	0,765 0,776	19,4 19,7
13/64	0,2031	5,2	0,202 0,208	5,1 5,3	25/32	0,7812	19,8	0,780 0,791	19,8 20,1
1/4	0,2500	6,3	0,249 0,255	6,3 6,5	7/8	0,8750	22,2	0,874 0,885	22,2 22,5
F	0,2570	6,5	0,256 0,263	6,5 6,7	57/64	0,8906	22,6	0,890 0,901	22,6 22,9
17/64	0,2656	6,7	0,265 0,272	6,7 6,9	29/32	0,9062	23,0	0,905 0,916	23,0 23,3
5/16	0,3125	7,9	0,311 0,318	7,9 8,1	1	1,0000	25,4	0,999 1,010	25,4 25,7
0	0,3160	8,0	0,315 0,322	8,0 8,2	1 1/64	1,0156	25,8	1,015 1,028	25,8 26,1
21/64	0,3281	8,3	0,327 0,334	8,3 8,5	1 1/32	1,0312	26,2	1,030 1,043	26,2 26,5

¹ Centering-holes (guide-holes) dimensioned with ϕ 2,1mm on drawings may be drilled (preferably) with ϕ 2,5mm, since the manufacturing cost justifies the alteration.

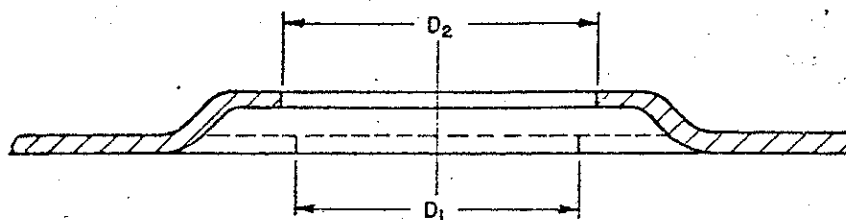
² It shall be applied the following tolerances for nominal dimensions not included on Table 4:

- a) $+0,1$
0 mm for nominal dimension ≤ 3 mm;
- b) $+0,2$
0 mm for $3 \text{ mm} < \text{nominal dimension} \leq 18$ mm; and
- c) $+0,3$
0 mm for $18 \text{ mm} < \text{nominal dimension}$.

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5.4.4 END HOLES

The tolerances for the end holes, obtained in parts with lightening holes or other similar circumstances, where an existing hole (± 0.5 mm) is reamed after sheet forming, shall be ± 1.0 mm. See Figure 14.



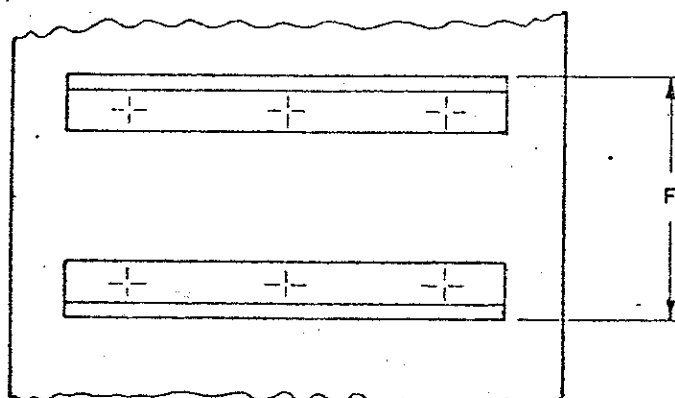
Hole end tolerance $D_2 = \pm 1.0$ mm

Existing hole tolerance $D_1 = \pm 0.5$ mm

FIGURE 14: LIGHTENING HOLE TOLERANCES

5.5 DIMENSIONS IN ASSEMBLIES

The tolerances for dimensions established in assemblies shall be within a ± 1 mm limit, unless otherwise indicated in the drawing. See Figure 15.



Tolerances for $F = \pm 1$ mm

FIGURE 15: TOLERANCES FOR ASSEMBLIES

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5.6 GRAIN DIRECTION

Grain direction tolerance (general or specific) is applicable to sheet metal or milled parts, manufactured from rolled, forged, or extruded raw material. In the application of grain direction, one of the following instances may occur:

- a. the drawing does not indicate grain direction but calls out GN 251 (see item 6) for parts on which the manufacture exerts influence upon grain direction. In this case, grain direction is optional, and shall not be considered for manufacture;
- b. grain direction is indicated by the production drawing, but with no specific angular tolerance. Example:

← GRAIN DIRECTION →

In this case, a general angular tolerance equal to $\pm 0.26 \text{ RAD } (\pm 15^\circ)$ shall be applied;

- c. grain direction and specific angular tolerance are indicated by the production drawing. Example:

← GRAIN DIRECTION: $\pm 0.17 \text{ RAD } (\pm 10^\circ)$ →

In this case, the indicated specific tolerance is mandatory.

6. GENERAL NOTES

6.1 DRAWING CALL OUT

This Standard shall be called out in the production drawings, by the note NI-090-TOLERANCES according to NE 03-007 and as indicated (see 6.5.2 and 1 of Figure 1(a)).

TECHNICAL STANDARD

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Tolerances, when they are standardized by this document, shall be called out in the Drawing Title Block, as specified in Figure 16, below.

UNLESS SPECIFIED IN DWG
ALL DIMENSIONS ARE IN MM.
TOLERANCE IN: NE 03-007

X	X,X	X,XX	ANG.
+	+	+	+
-	-	-	-

FIGURE 16: EXAMPLE OF STANDARD TOLERANCE CALL OUT

Tolerances which are not standardized by this document, but are unified, shall be called out in the Drawing Title Block, as exemplified in Figure 17, on the next page.

Unified tolerance means that there is a definite tolerance value for each group of dimensions that have the same number of digits after the point. For instance, if the unified tolerance for the group of dimensions of up to tenths of millimeters is ± 0.4 mm and the tolerance for the group of dimensions represented by whole millimeters is ± 1 mm (in the same drawing) the call out of these tolerances shall be as per Figure 17.

UNLESS INDICATED IN DWG:
-ALL DIMENSIONS ARE IN MM.
-TOLERANCE IN:

X	X,X	X,XX	ANG.
$\begin{smallmatrix} + \\ - \end{smallmatrix} 1$	$\begin{smallmatrix} + \\ - \end{smallmatrix} 0.4$	$\begin{smallmatrix} + \\ - \end{smallmatrix}$	$\begin{smallmatrix} + \\ - \end{smallmatrix}$

FIGURE 17: EXAMPLE OF UNIFIED, NON-STANDARD TOLERANCE CALL OUT

Tolerances which are neither unified nor standardized (as can be seen in item 6.1.2) shall be called out in the drawing itself, beside the nominal dimensions they apply to, as specified in Figure 18 below.

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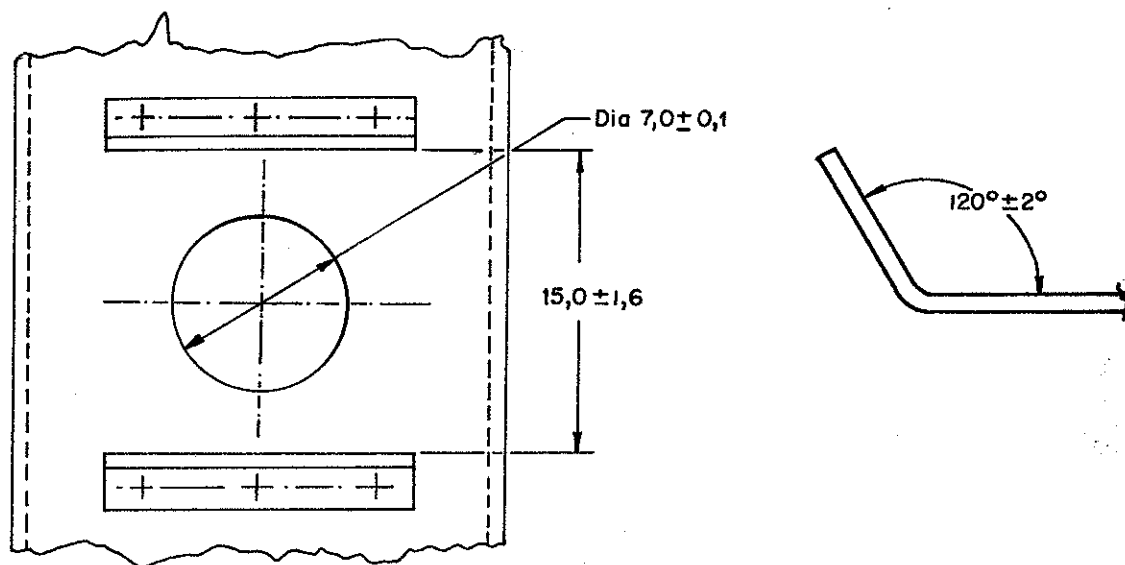


FIGURE 18 : EXAMPLES OF TOLERANCES INDICATED IN THE DRAWING

NOTE : In case the tolerance values are indicated in the drawing itself, those values shall be followed during the inspection, instead of those standardized by this document.

6.5 STANDARD CANCELLATION AND SUPERSEDURE

6.5.1 This Standard supersedes PARN 11003

6.5.2 This Standard had been deactivated by NE 03-063 Standard on July 16, 1985 (OE NOR 030393), for new programs.

For PROGRAMS IN COURSE, NE 03-063 Standard, shall be called out whenever possible.

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